Subsystem Power Requirements

EPS

Component	Voltage (V)	Current (mA)	Power (mW)
Solar Array	~7.4	0.0	0.0
Reverse Polarity Protection	7.4	5.0	37.0
MPPT	7.4	50.0	370.0
Battery Charger	7.4	30.0	222.0
Battery + Battery Protection Circuit	7.4	5.0	37.0
DC Bus	7.4	0.0	0.0
Buck Converters	5.0 / 3.3	35.0	223.0
Fuses	-	0.0	0.0
Load Switches	3.3	8.0	26.4

TT&C and OBC

Component	Voltage (V)	Current (mA)	Power (mW)
On Board Computer (RPi-CM4+)	5.0	400.0	2000.0
RF Switch	5.0	50.0	250.0
STM32 F4	3.3	80.0	264.0
Transceiver	3.3	80.0	264.0
Power Amplifier	5.0	600.0	3000.0
Low Noise Amplifier	3.3	10.0	33.0
Filter	-	0.0	0.0
Balun	-	0.0	0.0
Antenna	-	0.0	0.0

ADCS

Component	Voltage (V)	Current (mA)	Power (mW)
STM32 H7	3.3	80.0	264.0
Magnetometer	3.3	1.5	4.9
Sun Sensor	3.3	4.0	13.2

Component	Voltage (V)	Current (mA)	Power (mW)
Earth Sensor	3.3	15.0	49.5
GPS Receiver	3.3	25.0	82.5
Gyro Sensor	3.3	5.0	16.5
Magnetic Torquer	12.0	250.0	3000.0
Reaction Wheels	12.0	300.0	3600.0
Reaction Control System	12.0	500.0	6000.0

PAYLOAD

Component	Voltage (V)	Current (mA)	Power (mW)
STM32 F4	3.3	80.0	264.0
Optical Camera	5.0	300.0	1500.0
Solar Flux Sensor	3.3	5.0	16.5
Magnetometer	3.3	1.5	4.9
UV Sensor	3.3	5.0	16.5